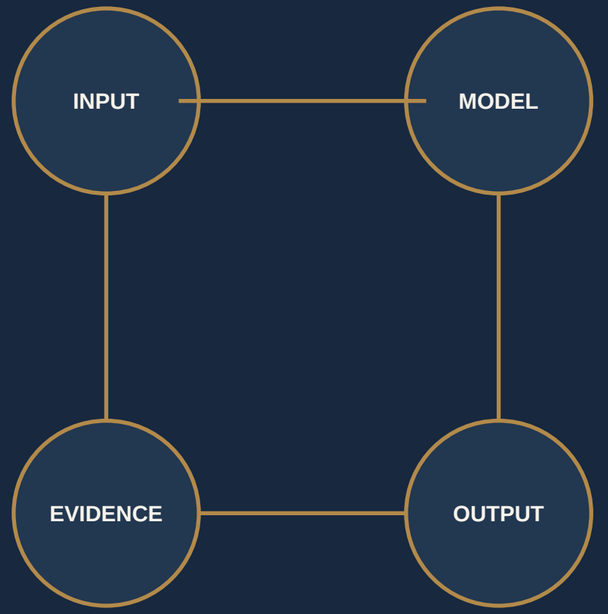
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Perception

AI.406 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define provide an advanced treatment of robot perception using vision.
- Apply key methods including lidar.
- Build or analyze a practical system involving depth sensors.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Provide an advanced treatment of robot perception using vision

02

METHODS

LiDAR

03

SYSTEMS

Depth sensors

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Provide an advanced treatmer

Provide an advanced treatment of robot perception using vision

02 · LiDAR

LiDAR

03 · Depth sensors

Depth sensors

04 · Tactile sensing

Tactile sensing

05 · Sensor fusion and learned per

Sensor fusion and learned perception

06 · Mandatory practical: Develop

Mandatory practical: Develop an integrated perception pipeline for an autonomous robot

07 · ### Year 4 — Semester 2

Year 4 — Semester 2

Core ideas, made explicit

AI.406 • Distinguish definitions, assumptions and evidence

- Provide an advanced treatment of robot perception using vision
- LiDAR
- Depth sensors
- Tactile sensing
- Sensor fusion and learned perception

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.406 • Distinguish definitions, assumptions and evidence

- Mandatory practical: Develop an integrated perception pipeline for an autonomous robot
- ### Year 4 — Semester 2

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM

Provide an advanced treatment of robot perception using vision

INPUTS

LiDAR

METHOD

Depth sensors

OUTPUT

Tactile sensing

EVALUATION

Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving provide an advanced treatment of robot perception using vision.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using provide an advanced treatment of robot perception using vision.
- Implement or simulate a baseline using lidar.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.